

EMAIL SPAM CLASSIFIER USING ARTIFICIAL INTELLIGENCE

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Project Title:

Email Spam Classifier using Artificial Intelligence

Objective:

The objective of this project is to classify emails as spam or not spam (ham) using Artificial Intelligence and Machine Learning techniques based on text data.

Tools and Technologies:

- Python
- Google Colab
- Scikit-learn
- Pandas
- NumPy

Dataset:

A sample email dataset containing labeled messages as spam or ham was used in this project. The dataset includes email text and corresponding classification labels.

Methodology:

Loaded the email dataset.

Converted text data into numerical features using CountVectorizer.

Split the dataset into training and testing sets.

Trained a Naive Bayes Classifier model.

Predicted whether emails are spam or not spam.

Evaluated model performance using accuracy and classification report.

Output:

- Accuracy: 100.0%
- Classification Report:

Precision: 1.00

Recall: 1.00

F1-score: 1.00

- Predicted Result: Spam

Result:

The Artificial Intelligence model successfully classified emails as spam with 100% accuracy on the test dataset. The Naive Bayes Classifier performed effectively in detecting spam messages.

Conclusion:

This project demonstrates how Artificial Intelligence and Natural Language Processing techniques can be used to classify email messages. The model successfully identified spam emails with high accuracy, showing the effectiveness of text classification algorithms.